

A248: MOT

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1 Abstract

A given magneto optical trap is adjusted to trap and cool an esamble of Rubidium atoms to a fraction of a Kelvin.

2 Theory

3 Setup

The setup consits of a laser part where the lasers are locked to the transition frequency for cooling and repumping and the main MOT part where three counterpropagating lasers cool and trap the atoms through the Zeeman shift of the magnetic field from colis in an anti-helmholz configuration (see Figure 1). Two lasers are set to the transition frequency

4 Measurements

5 Conclusion

References

- [1] Fp experiment: Rubidium mot tutorial. Lab manual, University of Bonn (January 2014). Dated January 2014

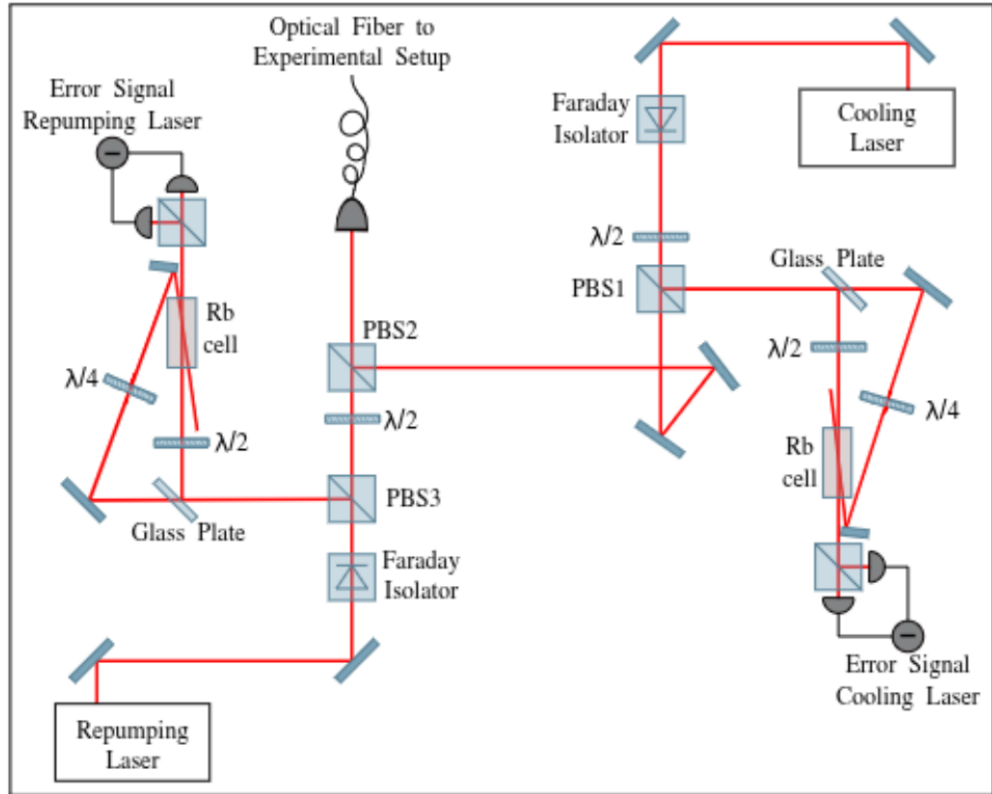


Figure 1: Laser part to lock the frequency to the repumping and cooling transition through polarisation spectroscopy.[1]